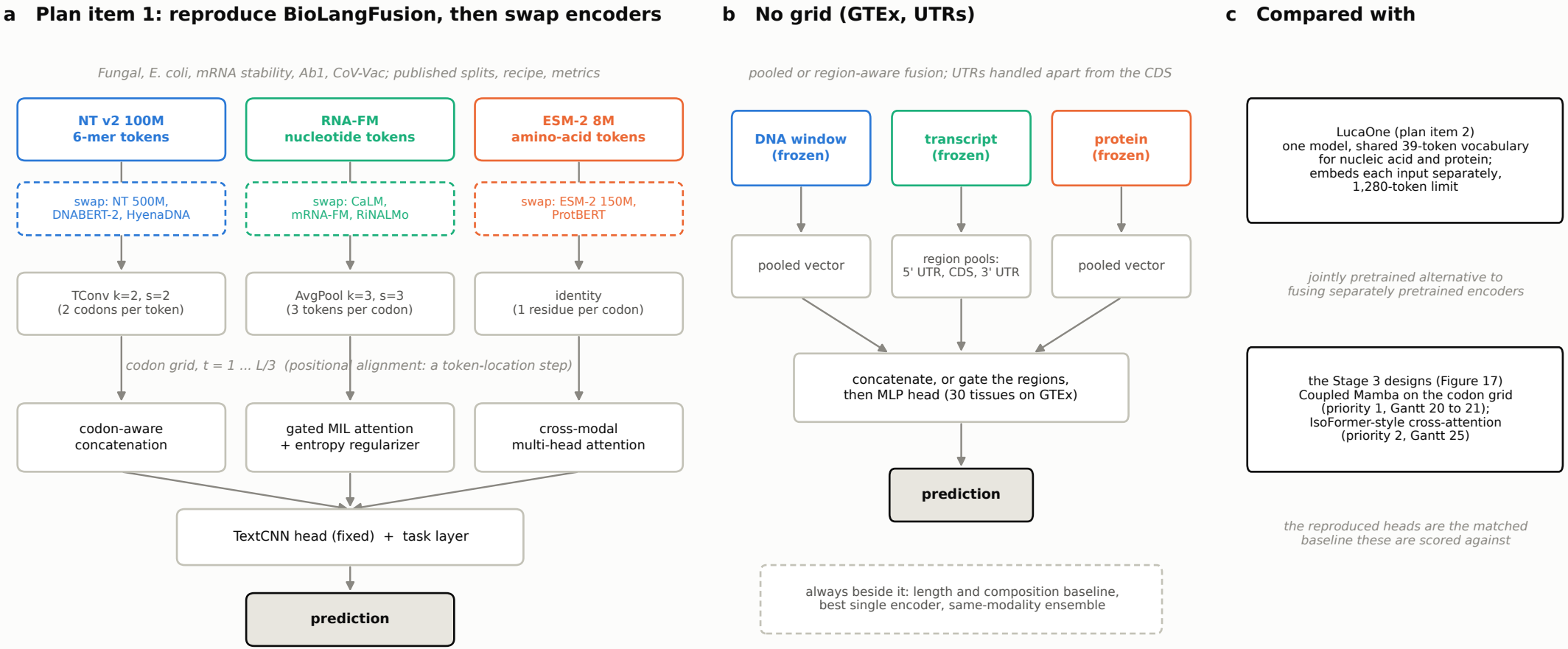
a Plan item 1: reproduce BioLangFusion, then swap encoders
Fungal, E. coli, mRNA stability, Ab1, CoV-Vac; published splits, recipe, metrics
NT v2 100M 6-mer tokens
RNA-FM nucleotide tokens
ESM-2 8M amino-acid tokens
swap: NT 500M, DNABERT-2, HyenaDNA
swap: CaLM, mRNA-FM, RiNALMo
swap: ESM-2 150M, ProtBERT
TConv k=2, s=2 (2 codons per token)
AvgPool k=3, s=3 (3 tokens per codon)
identity (1 residue per codon)
codon grid, t = 1 ... L/3 (positional alignment: a token-location step)
codon-aware concatenation
gated MIL attention + entropy regularizer
cross-modal multi-head attention
TextCNN head (fixed) + task layer
prediction
b No grid (GTEx, UTRs)
pooled or region-aware fusion; UTRs handled apart from the CDS
DNA window (frozen)
transcript (frozen)
protein (frozen)
pooled vector
region pools: 5' UTR, CDS, 3' UTR
pooled vector
concatenate, or gate the regions, then MLP head (30 tissues on GTEx)
prediction
always beside it: length and composition baseline, best single encoder, same-modality ensemble
c Compared with
LucaOne (plan item 2) one model, shared 39-token vocabulary for nucleic acid and protein; embeds each input separately, 1,280-token limit
jointly pretrained alternative to fusing separately pretrained encoders
the Stage 3 designs (Figure 17) Coupled Mamba on the codon grid (priority 1, Gantt 20 to 21); IsoFormer-style cross-attention (priority 2, Gantt 25)
the reproduced heads are the matched baseline these are scored against